

Lecture 4 - FIR Systems and Convolution

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November 10, 2025

- The Kronecker-Delta function is defined as:

$$\delta(n - m) = \begin{cases} 1, & \text{if } n = m \\ 0, & \text{otherwise} \end{cases}$$

- The impulse response $h(n)$ of a system is defined as the system's output, $y(n)$, when its input, $x(n)$, is $\delta(n)$.
- As computed in the previous lecture, for system's described by its (A, B, C, D) representation, its impulse response is given by:

$$h(n) = \begin{cases} 0, & \text{if } n < 0 \\ D, & \text{if } n = 0 \\ CA^{n-1}B, & \text{if } n > 0 \end{cases}$$

- The *convolutional equation* is a formula used to determine what the output of a system will be for a generic input, $x(n)$, based on its impulse response.
- The convolutional equation is given by two equivalent formulations:

$$y(n) = \sum_{k=-\infty}^{\infty} (h(n - k)x(k))$$
$$y(n) = \sum_{k=-\infty}^{\infty} (h(k)x(n - k))$$

- The specific equation which should be used is entirely a matter of convenience based on the problem at hand.
- Convolution is also a process used in continuous systems, but given by a different formula which will be covered in a future lecture.

Example 1

Using the general state-space equations and the definition of $h(n)$, derive the convolutional equation.

Solution:

For any input $x(n)$, out state space equations look like:

$S(n)$	$y(n)$
$S(0) = \mathbf{0}$	$y(0) = \mathbf{C}S(0) + \mathbf{D}x(0) = \mathbf{x}(0)$
$S(1) = \mathbf{A}S(0) + \mathbf{B}x(0) = \mathbf{B}x(0)$	$y(1) = \mathbf{C}S(1) + \mathbf{D}x(1) = \mathbf{C}Bx(0) + \mathbf{D}x(1)$
$S(2) = \mathbf{A}S(1) + \mathbf{B}x(1) = \mathbf{A}Bx(0) + \mathbf{B}x(1)$	$y(2) = \mathbf{C}S(2) + \mathbf{D}x(2) =$ $\mathbf{C}(\mathbf{A}Bx(0) + \mathbf{B}x(1) + \mathbf{B}x(1)) + \mathbf{D}x(2)$
$S(3) = \mathbf{A}S(2) + \mathbf{B}x(2) =$ $\mathbf{A}(\mathbf{A}Bx(0) + \mathbf{B}x(1)) + \mathbf{B}x(2)$	$y(3) = \mathbf{C}S(3) + \mathbf{D}x(3) =$ $\mathbf{C}(\mathbf{A}(\mathbf{A}Bx(0) + \mathbf{B}x(1)) + \mathbf{B}x(2)) + \mathbf{D}x(3)$
$\vdots$	$\vdots$
$S(n) = \sum_{k=0}^{n-1} \mathbf{A}^{n-1-k} \mathbf{B}x(k)$	$y(n) = \sum_{k=0}^{n-1} (\mathbf{C} \mathbf{A}^{n-1-k} \mathbf{B}x(k)) + \mathbf{D}x(n).$

From our earlier definition of $h(n)$, we know that $h(n) = \begin{cases} 0, & \text{if } n < 0 \\ \mathbf{D} & \text{if } n = 0 \\ \mathbf{C} \mathbf{A}^{n-1} \mathbf{B}, & \text{if } n > 0 \end{cases}$, and therefore $\mathbf{C} \mathbf{A}^{n-1-k} \mathbf{B}x(k) = h(n-k)$, and since $\mathbf{D} = h(0)$, we have:

$$y(n) = \sum_{k=1}^{n-1} h(n-k)x(k) + h(0)x(0) = \sum_{k=0}^{n-1} h(n-k)x(k)$$

Since, in the context of FIR systems, $h(n-k) = 0, \forall n < k$, and $\exists N \in \mathbb{N}$ such that $\forall n \geq N, h(n-k) = 0$, we have:

$$y(n) = \sum_{k=-\infty}^{\infty} h(n-k)x(k)$$

- The *effect* of a system is defined as
- The convolutional equation tells us how the effect of a system evolved over time.

Example 2

Find the output of the system $y(n) = \frac{1}{2}x(n) + \frac{1}{2}x(n-1)$ for the input $x(n) = \begin{cases} 1, & \text{if } n = 0, 2 \\ 2, & \text{if } n = 1 \\ 0, & \text{otherwise} \end{cases}$.

Solution:

From a previous question, we determined the impulse response of this system to be

$h(n) = \begin{cases} \frac{1}{2}, & \text{if } n = 0, 1 \\ 0, & \text{otherwise} \end{cases}$. Thus, using the convolutional equation, we can see that:

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} h(n-k)x(k) \\ &= \sum_{k=0}^{k=2} h(n-k)x(k) \\ &= h(n)x(0) + h(n-1)x(1) + h(n-2)x(2) = h(n) + 2h(n-1) + h(n-2) \end{aligned}$$

Therefore, we can determine that:

$$\begin{aligned} y(0) &= h(0) + 2h(-1) + h(-2) &= \frac{1}{2} \\ y(1) &= h(1) + 2h(0) + h(-1) &= \frac{3}{2} \\ y(2) &= h(2) + 2h(1) + h(0) &= \frac{3}{2} \\ y(3) &= h(3) + 2h(2) + h(1) &= \frac{1}{2} \\ y(4) &= h(4) + 2h(3) + h(2) &= 0 \\ \vdots & & \vdots \end{aligned}$$

Example 3

Use the impulse response $h(n) = \frac{1}{2}\delta(n) + \frac{1}{2}\delta(n-1)$ to derive the system $y(n) = \frac{1}{2}x(n) + \frac{1}{2}x(n-1)$.

Solution:

Using the convolutional equation, we can find that:

$$\begin{aligned}y(n) &= \sum_{k=-\infty}^{\infty} (h(n-k)x(k)) \\&= \sum_{k=-\infty}^{\infty} ((\frac{1}{2}\delta(n-k) + \frac{1}{2}\delta(n-k-1))x(k)) \\&= \sum_{k=-\infty}^{\infty} (\frac{1}{2}\delta(n-k)x(k) + \frac{1}{2}\delta(n-k-1)x(k)) \\&= \frac{1}{2}\delta(0)x(n) + \frac{1}{2}\delta(0)x(n-1) \\&= \frac{1}{2}x(n) + \frac{1}{2}x(n-1)\end{aligned}$$

- The identity system is defined as a system where the impulse response, $h(n)$, is the Kroncker-Delta function, $\delta(n)$.

Example 4

Find the output of the system $y(n) = x(n) + \alpha y(n-1)$ for the input $x(n) = 1$ using its impulse response.

Solution:

From a previous example, we know that the system's impulse response is given by:

$$h(n) = \begin{cases} \alpha^n, & \text{if } n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Therefore, using the convolutional equation, we find:

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} (h(k)x(n-k)) \\ &= \sum_{k=-\infty}^{\infty} (\alpha^k x(n-k)) \\ &= \sum_{k=0}^n (\alpha^k) \\ &= \frac{1 - \alpha^{n+1}}{1 - \alpha} \end{aligned}$$

- A *non-causal* system is a system whose current output depends on future inputs.
 - Image and audio systems are often implemented as non-causal systems.